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Green Task Scheduling for Energy-Efficient Cloud Computing

CSA0620 – Design and Analysis of Algorithms

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ABSTRACT

- This project develops a Green Task Scheduling system to reduce estimated energy consumption in cloud computing.
- Computational tasks are assigned to virtual machines based on their processing capacity and workload.
- Three scheduling algorithms, FCFS, Round Robin, and Green Scheduling, are implemented for comparison.
- The system calculates estimated energy consumption, task completion time, and virtual machine utilization.
- A web dashboard displays scheduling results and helps identify the most energy-efficient scheduling approach.

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INTRODUCTION

- Cloud computing provides virtual machines that execute computational tasks according to their processing requirements.
- Task scheduling determines which virtual machine should execute each incoming task.
- Traditional algorithms such as FCFS and Round Robin distribute tasks using predefined scheduling rules.
- Green task scheduling considers energy consumption along with processing capacity and task execution time.
- This project compares different scheduling algorithms to understand their energy consumption and overall performance.

OBJECTIVES

- To develop a Green Task Scheduling algorithm that assigns computational tasks to suitable virtual machines.
- To consider VM processing capacity, workload, estimated energy consumption, and execution time during task allocation.
- To estimate energy consumption using virtual machine power parameters and simulated CPU utilization.
- To evaluate scheduling performance using energy consumption, makespan, throughput, and CPU utilization.
- To compare Green Scheduling with FCFS and Round Robin using the same tasks and virtual machine configurations.

PROBLEM STATEMENT

- Traditional scheduling algorithms may assign tasks without considering the energy consumed by virtual machines.
- Uneven task distribution can overload some virtual machines while leaving other machines underutilized.
- Virtual machines with different processing capacities may require different amounts of time to complete the same task.
- Inefficient task allocation can increase estimated energy consumption and the time required to complete all tasks.
- Therefore, this project develops an energy-aware scheduling algorithm to improve task allocation and balance energy consumption with performance.

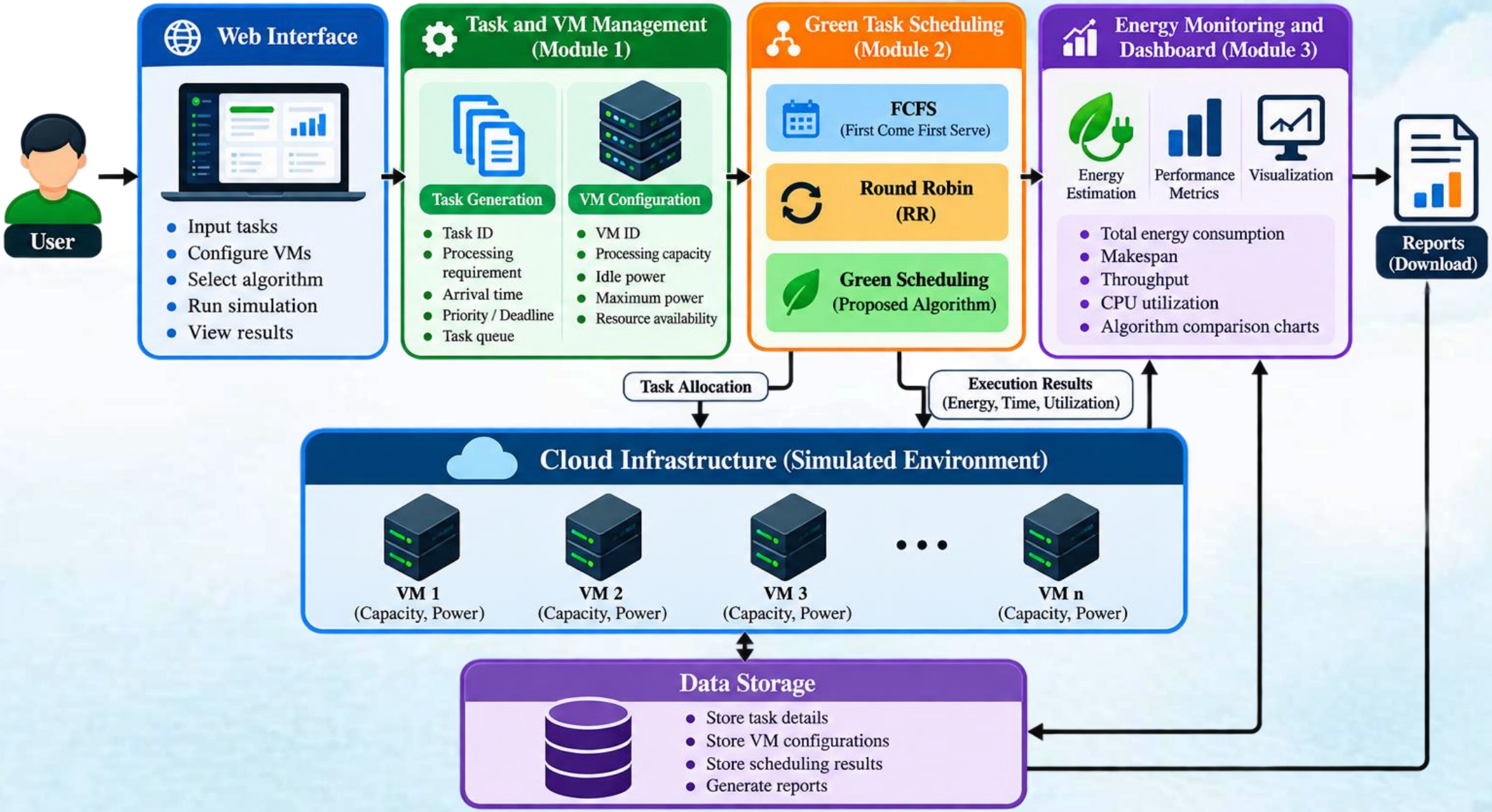
Literature Review

S. No.	Author & Year	Research Focus	Key Findings
1.	Beloglazov et al. (2012)	Energy-aware resource allocation	Improved energy efficiency through effective VM allocation and resource management.
2.	Beloglazov & Buyya (2012)	Dynamic VM consolidation	Reduced data center energy consumption while maintaining performance.
3.	Calheiros et al. (2011)	CloudSim simulation toolkit	Enabled simulation and evaluation of cloud resource allocation and scheduling algorithms.
4.	Buyya et al. (2009)	Cloud computing and resource management	Highlighted the importance of efficient resource utilization and scalable cloud services.

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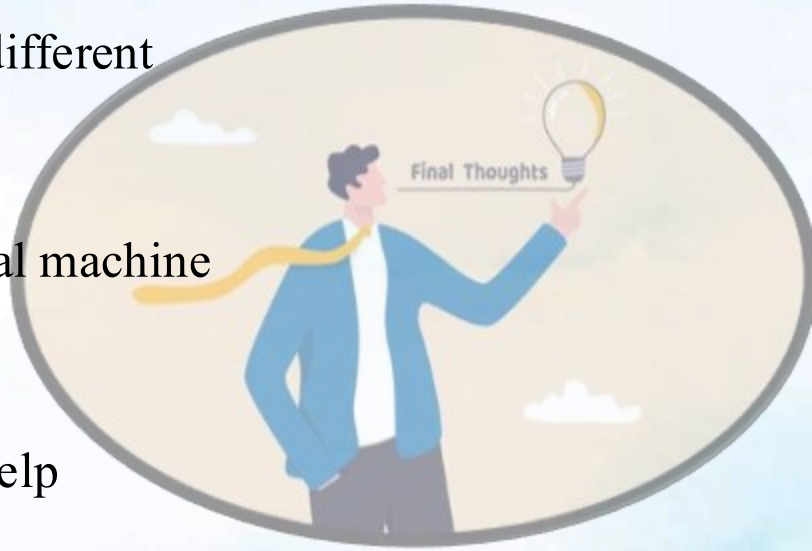
System Architecture

Green Task Scheduling for Energy-Efficient Cloud Computing



CONCLUSION

- The project provides a simulation-based system for evaluating energy-aware task scheduling in cloud computing.
- FCFS, Round Robin, and Green Scheduling are implemented to compare different approaches to task allocation.
- A configurable power model estimates energy consumption based on virtual machine utilization and execution duration.
- Performance metrics such as makespan, throughput, and CPU utilization help evaluate the scheduling algorithms.
- The web dashboard presents the simulation results clearly, allowing users to assess the trade-offs between energy consumption and performance.



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